



**UNIVERSITY  
OF ICELAND**

# **Meteorological Conditions and Rural Injury Accidents in Iceland**

Analysis of Accident Frequency and Traffic, 2007–2025

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M.Sc. thesis  
in Software Engineering



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30 ECTS thesis submitted in partial fulfilment of a  
*Master of Science* degree in Software Engineering

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# Abstract

This study examines how the occurrence of rural injury accidents varies with weather conditions in Iceland from 2007–2025. Mean wind speed is the primary weather measure, with wind gust and temperature as secondary measures. Accident records were linked to nearby 10-minute weather observations, and observed accident counts were compared with counts expected from local weather frequency. Two traffic-based extensions were then applied. The first approximately reweighted the O/E denominator using traffic responses observed at daily counters in 2019–2024. The second estimated accident rates per vehicle-kilometre in a counter-linked sample using observed daily traffic, rural road length, and pooled monthly weather frequencies. Matched-time and additional traffic models provided supporting checks.

Of 6,414 rural injury accidents, 6,259 had a qualifying wind observation within 20 km and five minutes of the recorded accident time. At mean wind speeds of at least 20 m/s, 68 accidents were observed compared with 31.6 expected, giving O/E 2.15. The approximate traffic correction increased this value to 2.71. In the separate 694-accident counter-linked analysis, the rate at 20 m/s or above was 0.667 accidents per million estimated vehicle-kilometres, 5.83 times the 0–5 m/s rate, although only 11 accidents occurred in the upper interval. Wind gust showed the same general upper-tail pattern, while temperature results were less consistent across methods.

Overall, rural injury accidents were over-represented during strong wind relative to local weather frequency. The traffic-based analyses pointed in the same direction but relied on additional exposure assumptions and smaller samples. The results describe statistical associations, not causal effects. A complete national estimate of accident risk per vehicle-kilometre would require hourly traffic data with broader spatial coverage.

# Ágrip

Í rannsókninni var kannað hvernig umferðarslys með meiðslum í dreifbýli á Íslandi tengjast veðurskilyrðum á árunum 2007–2025. Megináhersla var á meðalvindhraða, en vindhviður og hitastig voru einnig skoðuð. Slysagögn voru tengd við 10 mínútna veðurmælingar frá nálægum veðurstöðvum og skráður fjöldi slysa borinn saman við væntan fjölda miðað við staðbundna tíðni viðkomandi veðurskilyrða. Tvær viðbótargreiningar tóku síðan tillit til umferðar. Í þeirri fyrri var gerð áætluð umferðarleiðrétting á O/E-nefnaranum út frá breytingum í daglegri umferð á umferðarteljurum 2019–2024. Í þeirri síðari var slysátíðni á hvern ekinn kílómetra metin í minna úrtaki með daglegum umferðartölum, vegalengd og mánaðarlegri tíðni veðurs. Þöruð tímagreining og fleiri umferðargreiningar voru notaðar til stuðnings.

Af 6.414 slysum með meiðslum í dreifbýli var hægt að tengja 6.259 við gilda vindmælingu innan 20 km og fimm mínútna frá skráðum slysátíma. Við meðalvind 20 m/s eða meiri voru skráð 68 slys, en miðað við staðbundna tíðni slíks vinds mátti vænta 31,6 slysa. O/E-hlutfallið var því 2,15 og hækkaði í 2,71 eftir áætlaða umferðarleiðréttingu. Í sérstakri greiningu á 694 slysum með tengingu við umferðarteljara var áætluð slysátíðni við 20 m/s eða meiri 0,667 slys á hverja milljón ekinna kílómetra, eða 5,83-föld tíðnin við 0–5 m/s. Aðeins 11 slys voru í efsta vindbilinu. Sambærilegt mynstur kom fram við miklar vindhviður, en niðurstöður fyrir hitastig voru breytilegri eftir greiningaraðferð.

Í heild komu slys með meiðslum hlutfallslega oftar fyrir við mikinn vind en vænta mátti miðað við staðbundna tíðni vindsins. Umferðargreiningarnar bentu í sömu átt en byggðust á frekari forsendum um umferð og minni úrtökum. Niðurstöðurnar lýsa tölfræðilegum tengslum en sýna ekki fram á orsakasamband. Heildstætt mat á slysaáhættu á hvern ekinn kílómetra fyrir landið allt myndi krefjast umferðargagna á klukkustundargrunni með víðtækari landfræðilegri þekju.

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# Abbreviations

|      |                                                            |
|------|------------------------------------------------------------|
| ADU  | Average annual daily traffic.                              |
| CI   | Confidence interval.                                       |
| O/E  | Observed-to-expected accident ratio.                       |
| OR   | Odds ratio.                                                |
| RR   | Rate ratio.                                                |
| SDU  | Summer daily traffic, June–September.                      |
| VDU  | Winter daily traffic, December–March.                      |
| VHDU | Spring/autumn daily traffic derived from ADU, SDU and VDU. |
| VKT  | Vehicle-kilometres.                                        |



# 1 Introduction

## 1.1 Context and Motivation

Weather affects both driving conditions and the decision to travel. Strong wind can make a vehicle harder to control, low temperature can coincide with slippery roads, and darkness can reduce visibility. These conditions are relevant to Icelandic roads, where terrain can produce substantial local variation in wind (Snæbjörnsson et al., 2007; Pu et al., 2018). Precipitation and fog are also relevant road-weather hazards, but they are context for this study and are not analysed here.

Previous research provides two reasons to examine wind. Engineering studies describe mechanisms by which crosswind can destabilise vehicles, while observational studies compare accident occurrence with the frequency of windy conditions (Baker, 1986, 1994; Edwards, 1994). These are complementary forms of evidence: a physical mechanism does not establish how often accidents occur across a road network. Reviews also find less consistent road-safety evidence for wind than for precipitation (Theofilatos and Yannis, 2014). Local evidence is therefore needed to determine whether the same pattern is present on Icelandic roads.

In Iceland, wind-related road safety has been studied using vehicle-reliability analyses and operational vehicle-wind models (Snæbjörnsson et al., 2007; Hrafnkelsdóttir, 2020). This thesis addresses a different question: whether rural injury accidents are over-represented under particular weather conditions relative to how often those conditions occur locally. It then examines how the interpretation changes when available traffic information is introduced. The study therefore links a long Icelandic accident series with local weather frequency and two complementary traffic-exposure approaches, while treating station observations as proxies for road conditions rather than direct measurements of the forces acting on a vehicle.

The study covers rural injury accidents in Iceland from 2007–2025. Accident times and locations are linked to nearby 10-minute weather observations. Mean wind is the primary weather measure; wind gust and temperature are secondary measures, while season, time of day and daylight provide additional context.

Accident counts alone do not measure accident risk. Severe weather may increase the hazard faced by each vehicle while also reducing the number of vehicles on the road. Weather frequency and traffic volume are therefore considered separately (Andrey et al., 2003; Lord and Mannering, 2010; Washington et al., 2020).

## 1.2 Aim and Research Questions

The aim is to quantify how the occurrence of rural injury accidents varies with weather and what the available traffic data add to the mean-wind result. Three questions guide the analysis:

1. **Q1 — Primary weather-frequency O/E:** How does accident occurrence relative to local weather frequency vary across mean-wind, wind-gust and temperature intervals, overall and by season? The figures distinguish minor injury from serious or fatal injury; the combined all-injury result provides the main comparison across methods.
2. **Q2 — Approximate traffic correction:** How does the primary O/E change when its weather-frequency denominator is adjusted using the daily traffic response observed at counters in 2019–2024?
3. **Q3 — Daily-counter VKT rate:** How do injury-accident rates vary across mean-wind and gust intervals when estimated vehicle-kilometre exposure is constructed from observed daily traffic, rural road length, and pooled monthly weather frequencies?

The three questions form a deliberate sequence. Q1 gives the broad weather-frequency comparison for 2007–2025. Q2 asks how that result changes under an approximate traffic correction estimated from the available counters. Q3 then uses a smaller 2019–2024 counter-linked sample to construct rates per estimated vehicle-kilometre. Matched-time and additional annual and daily traffic models are retained as supporting robustness checks rather than separate main questions. Minor-injury and serious or fatal accidents together form the all-injury outcome. The study estimates associations, not causal effects or a complete national accident rate per vehicle-kilometre.

## 1.3 Thesis Outline

Chapter 2 gives the background for the weather and traffic comparisons. Chapter 3 describes the data, data preparation, and statistical methods. Chapter 4 presents the three main analyses in research-question order, followed by supporting checks. Chapter 5 discusses their interpretation and limitations, and Chapter 6 gives the conclusions. The appendix contains additional checks that support interpretation of the main results.

## 2 Background

### 2.1 Wind and Rural Road Accidents

Crosswind can produce overturning, lateral sideslip, or excessive yaw. The critical condition depends on vehicle geometry and mass, vehicle speed, road friction, wind direction, and driver response rather than wind speed alone (Baker, 1986, 1994). High-sided vehicles are particularly exposed, but the physical mechanisms also provide a reason to examine loss-of-control accidents more generally. These engineering studies establish physical plausibility and vehicle-specific failure mechanisms; they do not by themselves establish a population-level association between wind and accident occurrence.

This problem is directly relevant in Iceland, where terrain can accelerate and redirect near-surface wind over short distances. An Icelandic reliability study by Snæbjörnsson et al. (2007) modelled accident probability as a combined function of wind velocity and direction, road camber and friction, and vehicle speed. A station measures wind at one point, whereas vehicle stability also depends on the wind's direction relative to the road and on local road conditions. Terrain between a station and a road can alter the wind reaching the vehicle. Station observations therefore remain exposure proxies, and a shorter matching distance does not guarantee identical conditions. Mobile road measurement research similarly shows that roadside terrain creates marked spatial variation that a sparse fixed-station network cannot fully resolve (Pu et al., 2018).

Vegagerðin has developed a vehicle-wind model to support decisions about hazardous conditions and road closures, while also noting the limited experimental evidence available for passenger cars (Hrafnkelsdóttir, 2020). The present thesis does not test that operational model; it examines population-level accident occurrence using a separate data design.

Empirical work has compared the share of accidents occurring in high wind with the share of time that high wind was observed, an approach closely related to the observed/expected comparison used here (Edwards, 1994). Studies linking crashes to nearby weather stations likewise show that station observations can be informative while retaining measurement error (Young and Liesman, 2007). More broadly, Theofilatos and Yannis (2014) report less consistent road-safety evidence for wind than for precipitation.

### 2.2 Temperature, Daylight, and Time of Day

Temperature, daylight, and time of day add context that wind alone cannot provide. They affect road-surface conditions, visibility, travel patterns, and the composition of road users, but accident counts in these categories are not rates unless the corresponding amount of travel is known. Temperature effects can also be non-linear and season-specific: a Seoul time-series

study found its strongest cold-weather association below freezing in winter, while an hourly case-crossover study in Shenzhen found a J-shaped temperature relationship (Lee et al., 2014; Zhan et al., 2020). Temperature is therefore compared with its local seasonal frequency rather than treated as having one linear effect.

Darkness presents the same denominator problem. A method developed using Norwegian, Swedish, and Dutch data compared daylight and darkness within clock time and season, avoiding the interpretation of raw night-time counts as risk (Johansson et al., 2009). Here, raw hour, season, and daylight counts describe the sample. Matched daylight and adjusted severity models are used for numerical comparisons.

## **2.3 Traffic and Accident Rates**

Weather can change the amount of traffic as well as the hazard. Research using automatic traffic recorders shows that extreme weather variables, including wind, can reduce or redistribute hourly traffic volume (Sathiaraj et al., 2018). Accident counts during strong wind can remain low even if risk to each travelling vehicle is higher. High-resolution crash studies illustrate the value of hourly weather data rather than daily or monthly averages (Xing et al., 2019). The analyses distinguish three quantities: accident occurrence relative to weather frequency, rates based on estimated vehicle-kilometres, and changes in observed daily traffic. Studies with real-time traffic, weather, and road geometry show what a more complete analysis could achieve when such data are available (Ahmed et al., 2012).

## 3 Data and Methods

### 3.1 Data

#### 3.1.1 Source Data

Table 3.1 summarises the source datasets. Official documentation for each source is listed in the bibliography (Samgöngustofa, 2026; Veðurstofa Íslands, 2026; Vegagerðin, 2026; Hagstofa Íslands, 2026).

*Table 3.1: Source datasets. The study period is 2007–2025.*

| Data              | Provider                              | Contents                                                                         | Coverage                                                                   |
|-------------------|---------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Accident          | Samgöngustofa                         | Accident events and road links                                                   | 126,607 events and 126,610 road links, 2007–2025                           |
| Weather           | Icelandic Meteorological Office (IMO) | 10-minute weather observations and station metadata                              | 232,459,562 observations from 338 stations; 438 station records, 2007–2025 |
| Traffic and roads | Vegagerðin                            | Annual road traffic, daily counter reports, counter locations, and road geometry | 33,757 section-years, 2000–2025; 1,033,659 counter channel-days, 2019–2024 |
| Urban areas       | Statistics Iceland                    | Urban-area polygons                                                              | 97 polygons in the 2020–2024 release                                       |

Records from the year 2025 are included in the accident, injury, vehicle, road-link, annual-traffic and weather datasets under the same selection and quality criteria as records from earlier years. Daily counter data are available through 2024. Table 3.2 defines the weather and traffic quantities used in the subsequent comparisons.

*Table 3.2: Definitions used throughout the thesis.*

| Term  | Definition                                                                    | Use in this thesis                                                            |
|-------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| f     | 10-minute mean wind speed, m/s                                                | Main weather measure, matched within five minutes of the accident time.       |
| fg    | Wind gust reported in the same 10-minute observation, m/s                     | Additional weather measure; not a daily maximum.                              |
| t     | Air temperature reported with each 10-minute observation, °C                  | Additional weather measure matched independently.                             |
| O/E   | Observed accidents divided by accidents expected from local weather frequency | Main relative-occurrence result; traffic is not included.                     |
| ADU   | Average vehicles per day over the year                                        | Used to check and derive annual traffic quantities.                           |
| SDU   | Average vehicles per day in June–September                                    | Official summer traffic value.                                                |
| VDU   | Average vehicles per day in December–March                                    | Official winter traffic value.                                                |
| VH DU | Average vehicles per day in April–May and October–November                    | Spring/autumn daily traffic calculated as the residual from ADU, SDU and VDU. |

### 3.1.2 Accident Data

The accident dataset contains 126,607 records from 2007–2025 (Samgöngustofa, 2026). One record lacks valid temporal or spatial information, leaving 126,606 valid records.

Rural status is derived from accident coordinates and Statistics Iceland urban-locality polygons (Hagstofa Íslands, 2026). Points inside an urban polygon are classified as Urban, and valid points outside all polygons as Rural. A defined Vestmannaeyjar area is also classified as Urban. The same 2020–2024 polygon dataset is applied to the full 2007–2025 accident period. Changes in urban boundaries before 2020 are not represented.

Applying this rural classification yields 24,005 accidents, of which 6,414 involve injury. Of these, 6,259 have a qualifying primary weather match within 20 km and five minutes.

Accident severity is recorded in `meidsli`: code 1 denotes a fatal accident, code 2 a serious accident, code 3 a minor-injury accident, and code 4 an accident without injury. The main outcome is  $\text{meidsli} \leq 3$ . The serious or fatal comparison uses  $\text{meidsli} \leq 2$ . The source data also contain 17,591 rural damage-only accidents; these are outside the injury outcome.

For the reader-facing results, injury accidents are divided into two disjoint groups: minor injury ( $\text{meidsli}=3$ ) and serious or fatal injury ( $\text{meidsli}=1$  or  $2$ ). The O/E calculation for these groups is described in Section 3.3.1.

Figure 3.1 shows the geographic distribution of the rural injury sample and highlights accidents matched to strong mean wind.

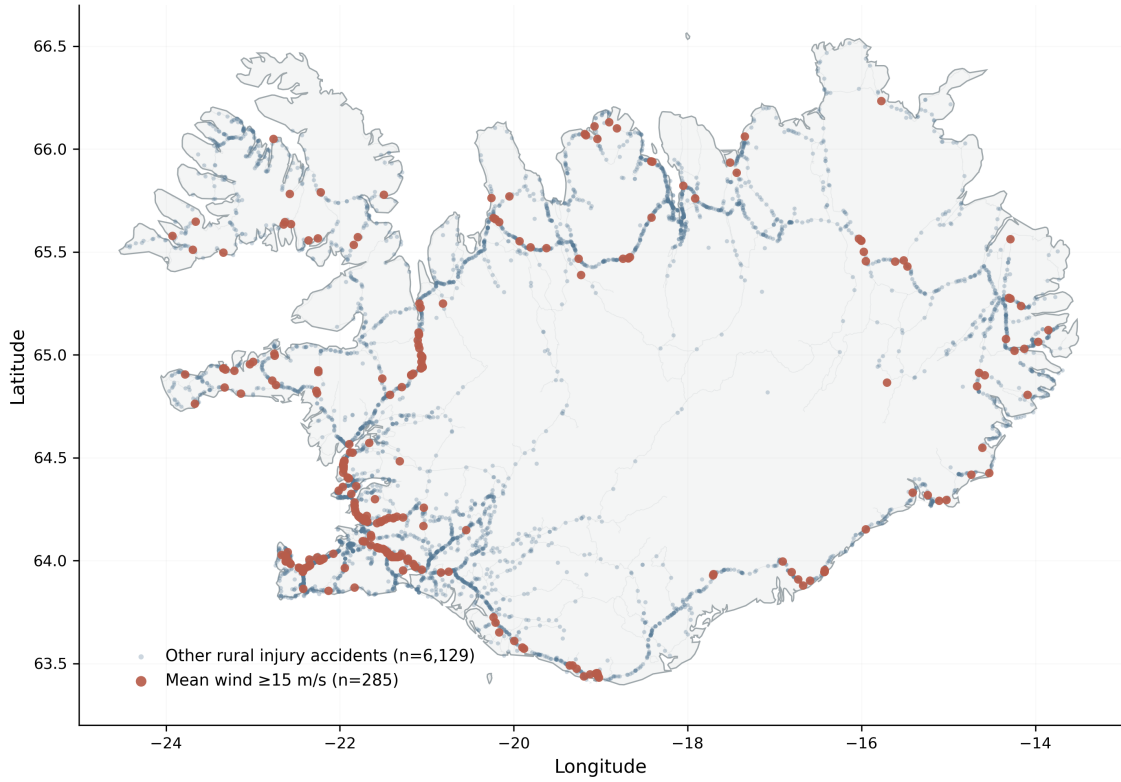


Figure 3.1: Locations of the 6,414 rural injury accidents. Accidents matched to mean wind of at least 15 m/s are highlighted. The map shows the spatial coverage of the study and is not a regional accident-rate map.

### 3.1.3 Weather Data

The weather dataset contains 10-minute observations from Icelandic weather stations. The variables used here are 10-minute mean wind speed ( $f$ ), reported wind gust ( $fg$ ), and temperature ( $t$ ). For each accident, the gust is taken from the same matched 10-minute observation as mean wind; it is not the maximum across the accident day. Mean wind speed remains the main weather measure. A wind observation is used when

$$0 \leq f < 45, \quad 0 \leq fg < 65, \quad fg + 0.5 \geq f.$$

Rows missing either  $f$  or  $fg$  are excluded. Negative values, values at or above either upper limit, and internally inconsistent values where  $fg = 0$  but  $f > 0$  are excluded. These quality exclusions are summarised in Table 3.4. A value of  $f = 0$  alone is kept. Only uninterrupted all-zero runs where  $f = fg = 0$  for at least two hours are excluded as potentially frozen sensors. Missing temperature does not affect whether a wind observation is used.

Each accident is matched to the geographically nearest station with a quality-controlled wind observation at one of the two adjacent timestamps on the 10-minute observation grid. The selected observation is therefore no more than five minutes from the recorded accident time. Because matching is based on absolute time difference, the adjacent observation can fall on the neighbouring calendar day for accidents occurring within five minutes of midnight. The maximum matching distance is 20 km. The matching dataset includes all 6,414 accidents and records whether each accident meets the weather-match criterion.

Temperature is matched independently within the same 20 km and five-minute limits. The nearest valid value between  $-30$  and  $30^{\circ}\text{C}$  may therefore come from a different station when the wind station lacks temperature. In the final analysis dataset, however, all 6,259 wind and temperature matches use the same station and observation time.

The IMO supplied observations of time, station, temperature, mean wind, and gust. The same quality-controlled weather data are used for accident matching, background frequencies, matched control times, and traffic analyses. No temperature value is imputed.

Table 3.3 reports spatial and temporal match quality for wind and temperature. Table 3.4 then summarises the wind-quality exclusions and retained observations.

*Table 3.3: Share of accidents with weather information from a station with a valid measurement within 20 km. Distance is shown as median / 90th percentile; every retained observation is within five minutes of the accident time.*

| Variable           | Accidents | Share | Stations | Distance (km) |
|--------------------|-----------|-------|----------|---------------|
| Mean wind and gust | 6,259     | 97.6% | 279      | 5.1 / 12.3    |
| Temperature        | 6,259     | 97.6% | 279      | 5.1 / 12.3    |

*Table 3.4: Wind-data scope and quality audit, 2007–2025. Quality-rule shares use all supplied observations. Every supplied station-year contains at least one wind measurement.*

| Category                                      | Records     | Share   |
|-----------------------------------------------|-------------|---------|
| All supplied station-time observations        | 232,459,562 | 100.00% |
| Missing f or fg                               | 261,519     | 0.11%   |
| Negative or upper-threshold wind              | 8,766       | 0.00%   |
| Internally inconsistent f/fg                  | 257,245     | 0.11%   |
| Frozen all-zero runs ( $\geq 2$ hours)        | 1,473,082   | 0.63%   |
| Quality-controlled wind observations retained | 230,458,950 | 99.14%  |

The wind-quality rules exclude 2,000,612 observations and retain 230,458,950 (99.14%). The largest excluded category is uninterrupted all-zero runs, which is reported separately because it reflects the rule for possible instrument stoppages rather than a missing observation. The audit reports each rule separately and uses all supplied weather observations from 2007–2025. For each accident’s station and season, the O/E expected count uses all quality-controlled background observations in that station–season group, not only observations recorded on accident days.

### 3.1.4 Traffic Data

Daily counter data underpin Q2 and Q3, while annual road-section traffic is used in supporting analyses.

Annual road-section data provide average annual daily traffic (ADU), summer daily traffic (SDU), winter daily traffic (VDU), section length, and annual vehicle-kilometres for 2000–2025 (Vegagerđin, 2026). The road-section comparison uses the years shared with the accident

study, 2007–2025. Following the official definitions, SDU covers June–September and VDU covers December–March. These data can be linked to accidents with a registered road-section identifier, but this linkage is considerably more restrictive than the weather matching. Table 3.5 distinguishes the published averages from the derived spring/autumn value.

*Table 3.5: Official traffic averages and the spring/autumn daily traffic value used in the annual-traffic analysis.*

| Period                                  | Months                         | Daily traffic reference                                                                     |
|-----------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------|
| ADU ( <i>ársdagsumferð</i> )            | Full calendar year             | Official annual daily traffic                                                               |
| VDU ( <i>vetrardagsumferð</i> )         | December–March                 | Official winter daily traffic                                                               |
| SDU ( <i>sumardagsumferð</i> )          | June–September                 | Official summer daily traffic                                                               |
| VHDU ( <i>vor- og haustdagsumferð</i> ) | April–May and October–November | Residual daily average calculated from ADU, SDU, VDU, and the number of days in each period |

Throughout the analyses, the four study seasons are aligned with these traffic periods: winter is December–March, spring is April–May, summer is June–September, and autumn is October–November. These are study-specific traffic-aligned seasons rather than conventional three-month meteorological seasons. Using June–September as summer keeps the weather comparison aligned with the Icelandic road authority’s summer traffic period; it is an operational definition, not a claim that weather is uniform across those months. The IMO also uses June–September when summarising the Icelandic summer (Veðurstofa Íslands, 2025). VH DU is not calculated as  $ADU - SDU - VDU$ . Annual traffic implied by ADU is first converted to a calendar-year total; the day-weighted SDU and VDU totals are subtracted, and the remainder is divided by the number of spring and autumn days:

$$VH DU = \frac{D_{year}ADU - D_{VDU}VDU - D_{SDU}SDU}{D_{VH DU}}.$$

Here, each  $D$  is the number of calendar days in that period. VH DU is therefore a residual estimate rather than an additional official traffic measure.

Daily traffic counts extracted from Vegagerðin reports cover 2019–2024 and provide observed totals at selected counter sites. They underpin the traffic response used in Q2 and the counter-linked exposure calculation in Q3, but do not provide hourly traffic at accident locations. More than one counter may occur on a road section, and counter positions are estimated from the registered road geometry. These data remain separate from the primary O/E calculation.

Of 68,946 annual road-section–year–period records, 61,442 have nearby usable wind data; separately, 763,171 of 774,274 daily counter-days have a matched daytime wind summary. The smaller linked accident samples used by the traffic models are reported in Chapter 4.

### 3.1.5 Selection Summary

Table 3.6 summarises the main data reductions. Selection for the study area and injury outcome is distinguished from invalid or missing measurements: an excluded record is not necessarily an erroneous record. The monthly-frequency VKT analysis starts with the 1,863 rural injury accidents in 2019–2024; the broader weather and annual-traffic samples cover

2007–2025. Traffic-period rows and counter-days are exposure records rather than accident counts, so the rows in the table are separate selections rather than one successive filtering chain.

*Table 3.6: Overview of the main data selections. Rows are separate selections rather than successive steps; the last three count accidents.*

| Dataset / selection          | Before      | Removed   | Retained    | Reason                                                                      |
|------------------------------|-------------|-----------|-------------|-----------------------------------------------------------------------------|
| Accidents                    | 126,607     | 120,193   | 6,414       | Invalid time/location, urban, or no injury.                                 |
| Weather observations         | 232,459,562 | 2,000,612 | 230,458,950 | Missing, implausible, inconsistent or frozen wind.                          |
| Annual traffic periods       | 68,946      | 9,072     | 59,874      | No usable weather or nonpositive traffic/length.                            |
| Daily counter-days           | 774,274     | 11,103    | 763,171     | No usable daytime wind summary.                                             |
| Primary weather sample       | 6,414       | 155       | 6,259       | No weather match within 20 km and five minutes.                             |
| Annual traffic sample        | 6,414       | 1,289     | 5,125       | Road linkage, exposure or weather eligibility.                              |
| Monthly-frequency VKT sample | 1,863       | 1,169     | 694         | Counter linkage, daytime accident, traffic record, or weather availability. |

### 3.1.6 Descriptive Accident Patterns

Figure 3.2 shows when the selected accidents occurred and the conditions recorded with them. The blue and red groups are the same separate severity groups used in the O/E figures. These are raw counts: more accidents in an hour or season may reflect more traffic, and the panels do not estimate accident risk. Vehicle counts and accident-type composition are shown in Appendix A.1.

## 3.2 Data Preparation and Reproducibility

Table 3.7 summarises the path from source data to the three main analyses. Accident, weather and traffic records first undergo quality checks and selection. Qualifying accidents are matched to nearby weather observations, and local weather frequencies form the denominator for Q1. Q2 applies an approximate traffic correction estimated from the 2019–2024 counters to that same weather-frequency framework. Q3 uses the counter-linked 2019–2024 sample to estimate vehicle-kilometre exposure from observed daily traffic, road length and pooled monthly weather frequencies. Matched-time and additional traffic models are retained as supporting checks. This separation is important because the analyses use different exposure assumptions and eligible samples.

All cleaning, matching, analysis, and thesis-table generation are scripted. Generated tables are written from analysis outputs, and validation checks cover key sample counts, matching rules, and exposure conservation. This reduces manual transcription between the analysis and the thesis while keeping the main transformations auditable.

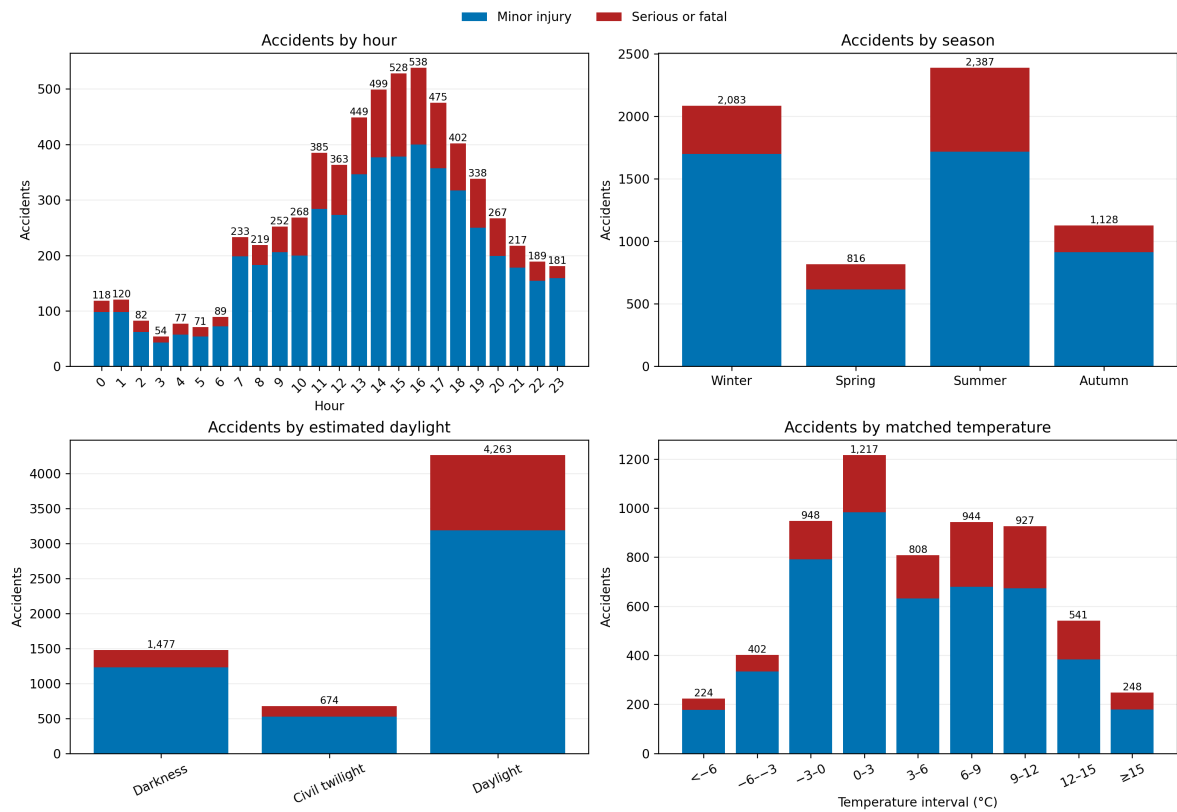


Figure 3.2: Rural injury accidents by hour, study season, estimated daylight and matched temperature. Blue shows minor injury (code 3), and red shows serious or fatal injury (codes 1–2). Labels give total accidents. Hour, season and daylight use 6,414 accidents; temperature uses 6,259 qualifying matches. These are descriptive counts, without traffic adjustment.

*Table 3.7: Data preparation and analysis workflow.*

| Stage                              | Main data                                                            | Output                                             | Role in thesis                            |
|------------------------------------|----------------------------------------------------------------------|----------------------------------------------------|-------------------------------------------|
| Data preparation                   | Accident, weather, road and traffic data                             | Quality-controlled datasets                        | Defines the study data.                   |
| Accident selection                 | Accident data and urban boundaries                                   | Rural injury accidents                             | Defines the study population.             |
| Weather matching                   | Accidents, weather observations and stations                         | Accident-time weather and local frequencies        | Common input to the weather analyses.     |
| Q1: weather-frequency O/E          | Accident weather and local frequencies                               | Observed and expected counts                       | Primary analysis.                         |
| Q2: approximate traffic correction | Primary O/E and 2019–2024 counter response                           | Traffic-corrected expected counts                  | Approximate traffic-exposure sensitivity. |
| Q3: monthly-frequency VKT rate     | Recorded counter-days, rural lengths and monthly weather frequencies | Accidents per estimated million VKT                | Main counter-linked traffic analysis.     |
| Supporting analyses                | Matched control times and alternative traffic panels                 | Matched odds ratios and alternative rate estimates | Robustness checks.                        |

## 3.3 Analysis

### 3.3.1 Q1: Weather-Frequency Comparison

Accident-time mean wind speed  $f$  is the main weather measure and is divided into 5 m/s intervals. It is the 10-minute mean wind observation matched within five minutes of the accident time, not an instantaneous measurement. Wind gust  $f_g$  from the same observation and independently matched temperature are additional weather measures. For each measure, accidents are compared with the distribution of that weather measure at the matched station in the same study season, pooled across all study years. Expected counts are therefore based on how often each condition occurred locally rather than on a national weather distribution. For example, if 5% of the relevant station–season observations fall in one wind interval, approximately 5% of that group’s accidents would be expected there under weather-frequency-proportional occurrence.

Let  $g$  identify a weather station and study season. Let  $A_{jg}$  be the number of accidents in weather interval  $j$  and group  $g$ , and let  $A_g = \sum_j A_{jg}$ . From all quality-controlled 10-minute observations, let  $T_{jg}$  be the number in interval  $j$  and  $T_g$  the total in group  $g$ . The local background frequency is

$$p_{jg} = \frac{T_{jg}}{T_g}.$$

If accident timing followed local wind frequency, the expected number in the interval would be  $A_g p_{jg}$ . The pooled observed/expected ratio is

$$R_j = \frac{\sum_g A_{jg}}{\sum_g A_g p_{jg}}.$$

A value of one means that the accident share equals the local time share of the weather interval. A value above one means that accidents are over-represented in that interval. Standardisation within station and season prevents locations or seasons with different weather distributions from dominating the result.

Table 3.8 gives a worked all-year example for mean wind of at least 20 m/s. For minor-injury accidents, 56 observed accidents are compared with 25.18 expected, giving O/E 2.22. For serious or fatal injury, 12 observed accidents are compared with 6.41 expected, giving O/E 1.87. The example illustrates the calculation; it does not imply that wind caused any individual accident.

*Table 3.8: Worked examples for the two O/E groups: mean wind at least 20 m/s, all years and seasons pooled.*

| Group                             | meidsli | Observed | Expected | O/E  |
|-----------------------------------|---------|----------|----------|------|
| Minor injury accidents            | 3       | 56       | 25.18    | 2.22 |
| Serious or fatal injury accidents | 1 or 2  | 12       | 6.41     | 1.87 |

The figures show the calculation separately for minor-injury accidents (*meiðsli* = 3) and serious or fatal accidents (*meiðsli* ≤ 2). These groups do not overlap. Each group's expected count uses its own accident totals within station and season. The combined all-injury O/E is also retained for comparison with the other analyses; it is calculated from summed observed and expected counts, not by averaging the two group ratios. Q1 is reported as a descriptive standardised ratio rather than a fitted inferential model, so observed and expected counts are shown without model-based confidence intervals. Wind gust and temperature are handled in the same way as mean wind. Temperature is matched independently, and its intervals are below −6, −6 to −3, −3 to 0, 0 to 3, 3 to 6, 6 to 9, 9 to 12, and at least 12°C. The O/E upper interval is combined at 12°C to keep the seasonal panels readable. The matched-time and annual-traffic models retain separate 12–15 and at-least-15°C intervals. Estimates across those different upper intervals are not treated as direct numerical comparisons.

### 3.3.2 Use of Traffic Data

The primary O/E denominator is the local frequency of wind conditions rather than traffic volume. Complete traffic data are unavailable at the required spatial and temporal resolution, so Q1 does not account for changes in travel volume during strong wind. If traffic is lower in strong wind while other factors remain unchanged, the accident-rate contrast per vehicle-kilometre would be larger than the weather-frequency O/E contrast.

Traffic enters the main analysis in two ways. Q2 estimates the relative daily traffic response across weather intervals from the 2019–2024 counters and uses that response to reweight the Q1 denominator. Q3 constructs estimated vehicle-kilometre exposure from observed daily traffic and rural road length in a smaller counter-linked sample. Neither analysis observes

hourly traffic. Additional annual road-section and allocated daily-counter models are retained as supporting checks under alternative exposure constructions.

### 3.3.3 Q2: Approximate Traffic Correction to O/E

The approximate correction uses the observed day-to-day traffic response at counter sections during 2019–2024. Among positive-count days, the mean count for the same counter section, year, month, and weekday defines expected daily traffic. Each day’s observed 24-hour count and expected count are allocated to weather intervals in proportion to their valid 07:00–24:00 weather minutes on that date. For interval  $j$ ,  $r_j$  is the sum of allocated observed traffic divided by the sum of allocated expected traffic. Zero-count days are excluded from this response calculation.

Using the same station–season group  $g$  and weather interval  $j$  as in Q1, the traffic-weighted weather frequency is

$$p_{jg}^* = \frac{p_{jg} r_j}{\sum_h p_{hg} r_h}.$$

Expected counts are then recalculated as  $A_g p_{jg}^*$ . Accident bins use weather at the recorded accident time, whereas  $r_j$  is based on a 24-hour traffic total repeated across the day’s daytime weather composition. Daily counters do not observe hourly traffic. The 2019–2024 response is also extrapolated to the 2007–2025 primary population. The result is therefore a model-based sensitivity analysis, not a directly observed traffic-adjusted O/E.

### 3.3.4 Q3: Monthly-Frequency Daily-Counter Rate

The main traffic rate is restricted to 2019–2024, when daily counter data are available. Eligible accidents must be rural injury accidents occurring between 07:00 and 24:00, have a counter section for the registered road and year, be successfully located on that section, have usable accident-time wind and gust within 20 km and five minutes, and have a traffic record on the accident date. For this analysis, the 20 km weather distance is measured from the counter-section location rather than from the accident coordinate. The same five-minute time rule applies as in Q1 and can select the adjacent observation across midnight. The denominator is broader than the accident set: it uses every eligible recorded counter-section day with a nonnegative traffic count and rural road length, including days without an accident. For counter section  $j$ , date  $d$ , and weather interval  $b$ , estimated exposure is

$$\hat{V}_{jdb} = C_{jd} L_j p_{j,m(d),b},$$

where  $C_{jd}$  is the observed 24-hour traffic count,  $L_j$  is rural section length under that year’s definition, and  $p_{j,m(d),b}$  is the 07:00–24:00 weather frequency for that calendar month at the station assigned to counter section  $j$ . The weather frequencies are pooled across 2007–2025 to provide stable station–month proportions, while both traffic and accident eligibility remain restricted to 2019–2024. The full daily count is allocated to the 07:00–24:00 window and the proportions sum to one, so exposure is conserved. Accidents are classified separately by observed event-time weather within 20 km and five minutes. The reported rate is

$$R_b = \frac{O_b}{\sum_j \sum_d \hat{V}_{jdb}} 10^6.$$

This construction uses actual day-to-day traffic variation and rural road length, but allocates it with a typical monthly weather distribution rather than observed hourly traffic.

### 3.3.5 Supporting Matched-Time Comparison

The supporting matched-time analysis uses a time-stratified case-crossover design in which each accident forms one matched set. The accident time is the case time, and the control times are the other occurrences of the same weekday and clock time in the same calendar month and year. For example, a Tuesday accident at 14:20 is compared with the other Tuesdays at 14:20 in that month. Mean-wind and gust controls use the accident's matched wind station; temperature controls use its independently matched temperature station. Control weather must satisfy the same five-minute observation-time rule, and both station matches are restricted to 20 km. Conditional logistic regression compares weather at the accident time with weather at its matched control times. This time-stratified referent strategy was selected to avoid overlap bias and to control calendar time without selecting controls from outside the accident month (Janes et al., 2005).

The model asks whether weather at an accident time differs from weather at comparable non-accident times. The matching holds station, year, month, weekday, and clock time fixed, but not unusual traffic, road conditions, or other factors that change on a particular day. Continuous effects are reported per 5 m/s or 5°C together with categorical estimates.

Mean wind and temperature are also entered together as categorical variables to assess whether the wind estimate changes after including temperature at the same case and control times. Daylight is calculated at the accident location for the accident date and the same matched dates. Because month, weekday, and clock time are held fixed, only strata that change between darkness, civil twilight, and daylight contribute information.

Seasonal differences in the mean-wind association are tested in a separate matched-time model. Mean wind is grouped as 0–10, 10–15, and at least 15 m/s for stability. The model includes interactions between wind interval and season, and a likelihood-ratio test compares it with a model having one common wind association across seasons. The omnibus test is used instead of judging seasonal differences from separate confidence intervals.

A separate logistic model describes serious or fatal rather than minor injury among accidents with complete wind and temperature matches. It includes broad categories of mean wind, temperature, daylight, time of day, and season. The result concerns injury severity among recorded accidents. It is not a model of whether an accident occurs and does not replace the traffic analyses.

### 3.3.6 Supporting Road-Section Traffic Comparison

Annual road-section data provide a supporting exposure analysis for accidents that can be linked by road section, year, official traffic period, and wind interval. For road-section–year–traffic-period stratum  $k$  and wind interval  $j$ , estimated vehicle-kilometres are

$$V_{kj} = q_k L_k D_k p_{kj},$$

where  $q_k$  is SDU, VDU, or derived VHDU vehicles per day,  $L_k$  is section length,  $D_k$  is the number of calendar days, and  $p_{kj}$  is the local share of quality-controlled 10-minute observations in wind interval  $j$ . This construction accounts for road length and differences in published seasonal traffic, but assumes that traffic within each traffic period is distributed across wind intervals in proportion to time. It does not observe traffic at the accident minute.

The station nearest the road-section midpoint with qualifying data supplies  $p_{kj}$ . Accidents in this analysis are matched to that road-section station, rather than to the accident-nearest station used in Q1, and are retained only when the station is within 20 km and has a valid observation within five minutes of the accident time.

A conditional Poisson model is fitted as a supporting inferential analysis. Each road section, year, and traffic period has its own intercept:

$$\log E(A_{kj}) = \alpha_k + \beta_j + \log V_{kj},$$

where  $A_{kj}$  is the accident count and  $\log V_{kj}$  is the offset. The exponentiated coefficient  $\exp(\beta_j)$  is the estimated rate ratio relative to 0–5 m/s within road-section–year–traffic-period strata. Strata with no accidents contain no within-stratum information after conditioning and do not contribute to this model. The model is fitted separately for all injury accidents and for serious or fatal accidents. A seasonal extension uses 0–10, 10–15, and at least 15 m/s because the finer upper-wind cells are sparse.

### 3.3.7 Supporting Allocated Daily-Counter Model

An additional daily-counter model uses observed 24-hour totals but estimates how those totals are distributed across wind intervals within each day. Directional channels are first summed to one count for each physical counter and date. Each 2019–2024 accident then requires a counter on the exact registered road section and year, positive traffic on the accident date, and a qualifying accident-time mean-wind observation. For counter  $c$ , date  $d$ , and wind interval  $j$ , the daily count is allocated as

$$\hat{Q}_{cdj} = Q_{cd} \frac{N_{cdj}}{N_{cd}},$$

where  $N_{cdj}$  is the number of valid 10-minute observations in interval  $j$  and  $N_{cd}$  is the total number of valid observations that day. Accidents are classified by accident-time mean wind from the same station used for the counter-day denominator. Estimated traffic and accidents are aggregated by counter, year, and wind interval, and a conditional Poisson model gives each counter–year its own intercept. The allocation assumes uniform traffic across valid observation times and is therefore an estimated within-day split rather than observed hourly exposure. Selection and sensitivity checks for this 762-accident supporting model are reported in the appendix.

## 4 Results

Results are presented in the order of the three research questions. First, the primary weather-frequency analysis compares observed accidents with counts expected from local weather frequency. Second, an approximate traffic correction asks how that comparison changes when the traffic response measured at daily counters is applied to the primary denominator. Third, the monthly-frequency counter analysis estimates accident rates per vehicle-kilometre in the smaller 2019–2024 traffic-linked sample. Matched-time and additional traffic models are reported afterwards as supporting checks. Because these analyses use different samples and denominators, their numerical estimates are not interpreted as repeated estimates of the same quantity.

### 4.1 Study Sample and Coverage

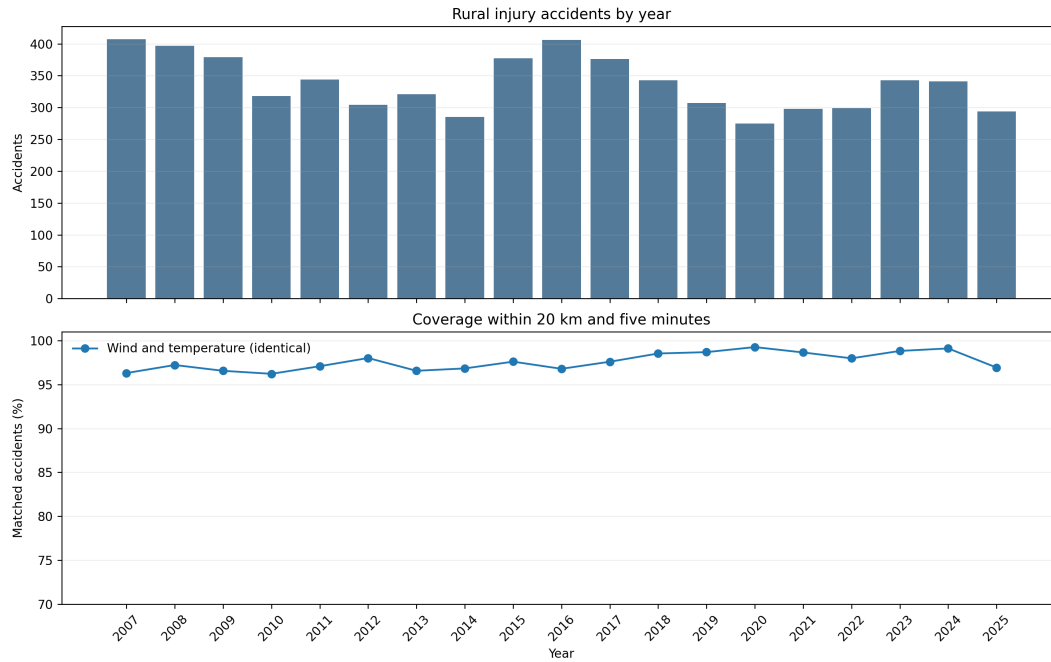
Table 4.1 summarises the populations used by the main and supporting analyses. The primary O/E retains 6,259 of the 6,414 rural injury accidents, whereas analyses requiring traffic linkage use smaller and more selective samples. Figure 4.1 shows that weather-match coverage remains high throughout 2007–2025.

*Table 4.1: Accident selection and analysis sample sizes.*

| Stage                              | Accidents | Purpose                                                                                    |
|------------------------------------|-----------|--------------------------------------------------------------------------------------------|
| Source records                     | 126,607   | Accident database, 2007–2025.                                                              |
| Rural injury accidents             | 6,414     | Study population.                                                                          |
| Q1: Primary O/E                    | 6,259     | Main weather-frequency analysis.                                                           |
| Q2: Approximate traffic correction | 6,259     | Same accident sample as Q1; denominator reweighted using counter-derived traffic response. |
| Q3: Monthly-frequency VKT          | 694       | Main counter-linked traffic analysis, 2019–2024.                                           |
| Supporting matched-time            | 6,257     | Time-matched robustness analysis.                                                          |
| Supporting annual traffic          | 5,125     | Broader road-linked traffic model.                                                         |
| Supporting allocated daily-counter | 762       | Alternative daily-counter model.                                                           |

The 20 km threshold provides broad coverage while limiting the distance between an accident and its weather proxy. It does not imply that station conditions are identical to conditions at the road, particularly in complex terrain. Wind and temperature are matched independently, but both procedures retain the same 6,259 accidents and select the same station and observation time in the final analysis dataset.

Question 3 uses a different starting population because daily counters are available only for 2019–2024. Table 4.2 shows the full selection from all injury accidents in 2007–2025, first restricting to 2019–2024 and then applying the Q3 linkage criteria, to the final 694 counter-linked daytime accidents. The first row includes urban and rural injury accidents; the



*Figure 4.1: Annual rural injury-accident counts and weather-match coverage, 2007–2025. The lower panel gives the percentage of accidents with a qualifying wind or temperature observation within 20 km and five minutes. Wind and temperature coverage is identical in every year, and no missing temperature value is imputed.*

rural restriction is applied in step (a). All 83 exclusions at the final step lack a counter-day record on the accident date; no accident is removed because its recorded daily count is zero.

*Table 4.2: Selection for the Q3 monthly-frequency daily-counter rate. The first row includes urban and rural injury accidents; the rural restriction is applied in step (a). The final set contains minor, serious and fatal injury accidents and is distinct from the supporting allocated daily-counter sample.*

| Step                                                         | Removed | Remaining |
|--------------------------------------------------------------|---------|-----------|
| All injury accidents, 2007–2025                              | –       | 17,045    |
| 0) Restrict to 2019–2024                                     | 11,849  | 5,196     |
| a) Exclude urban accidents                                   | 3,333   | 1,863     |
| b) Require counter-section for road/year                     | 994     | 869       |
| c) Require successful road-location assignment               | 21      | 848       |
| d) Exclude 00:00–07:00                                       | 67      | 781       |
| e) Require usable wind and gust ( $\leq 20$ km, $\pm 5$ min) | 4       | 777       |
| Require daily traffic count                                  | 83      | 694       |

*Notes:* The 21 road-location failures in step (c) comprise five records without usable road geometry, 15 located more than 100 m from their registered road, and one projected outside the counter sections. For step (e), the 20 km weather distance is measured from the counter-section location rather than the accident coordinate; valid mean wind and gust are required within five minutes of the recorded accident time.

## 4.2 Q1: Weather-Frequency O/E

Figures 4.2–4.4 show the primary weather-frequency comparisons for the full year and the four study seasons. Blue bars show minor-injury accidents and red bars show serious or fatal injury accidents; the groups do not overlap. The grey horizontal line marks  $O/E = 1$ , where observed accidents equal the number expected from local weather frequency. Values above one indicate over-representation relative to that frequency, not a measured increase in risk per vehicle-kilometre.

### 4.2.1 Mean Wind

Figure 4.2 shows the main result. The all-year O/E values are close to one at low and moderate mean wind speeds and rise in the upper intervals. At 20 m/s or above, 68 injury accidents were observed compared with 31.6 expected, giving the combined all-injury O/E of 2.15. The disjoint severity groups show the same broad upper-wind pattern: minor-injury accidents have O/E 2.22 and serious or fatal accidents O/E 1.87 in the highest interval. The latter group contains only 12 accidents, so the difference between the two bars is not evidence that severity itself changes with wind.

The seasonal panels are less stable than the pooled result. In particular, some of the largest seasonal O/E values are based on only a few accidents, so they do not support ranking seasons or injury groups. The 5 m/s intervals are used as descriptive bins rather than physical wind thresholds.

### 4.2.2 Wind Gust

Wind gust shows a similar upper-tail pattern in Figure 4.3. Across all injury accidents, 61 accidents were observed at gusts of at least 30 m/s compared with 18.2 expected, giving O/E 3.34. Both severity groups are also above one in this interval. Because gust and mean wind come from the same 10-minute observation and are strongly related measures, agreement between the two figures supports a general high-wind pattern rather than an independent replication.

### 4.2.3 Temperature

Temperature does not show the monotonic upper-tail pattern seen for wind. In Figure 4.4, O/E is below one around 3–6°C, while some colder and warmer intervals are above one. For minor-injury accidents, O/E is 0.72 at 3–6°C and 1.56 at 12°C or above. The serious or fatal group shows the same non-monotonic character, but several seasonal temperature extremes contain very few accidents. The pattern is therefore treated as secondary and is not interpreted as a causal temperature effect.

Adding year to the station–season standardisation changes the selected O/E estimates only slightly; for example, the all-injury mean-wind estimate at 20 m/s or above changes from 2.15 to 2.17. The pre-specified station–season comparison is therefore retained as Q1.

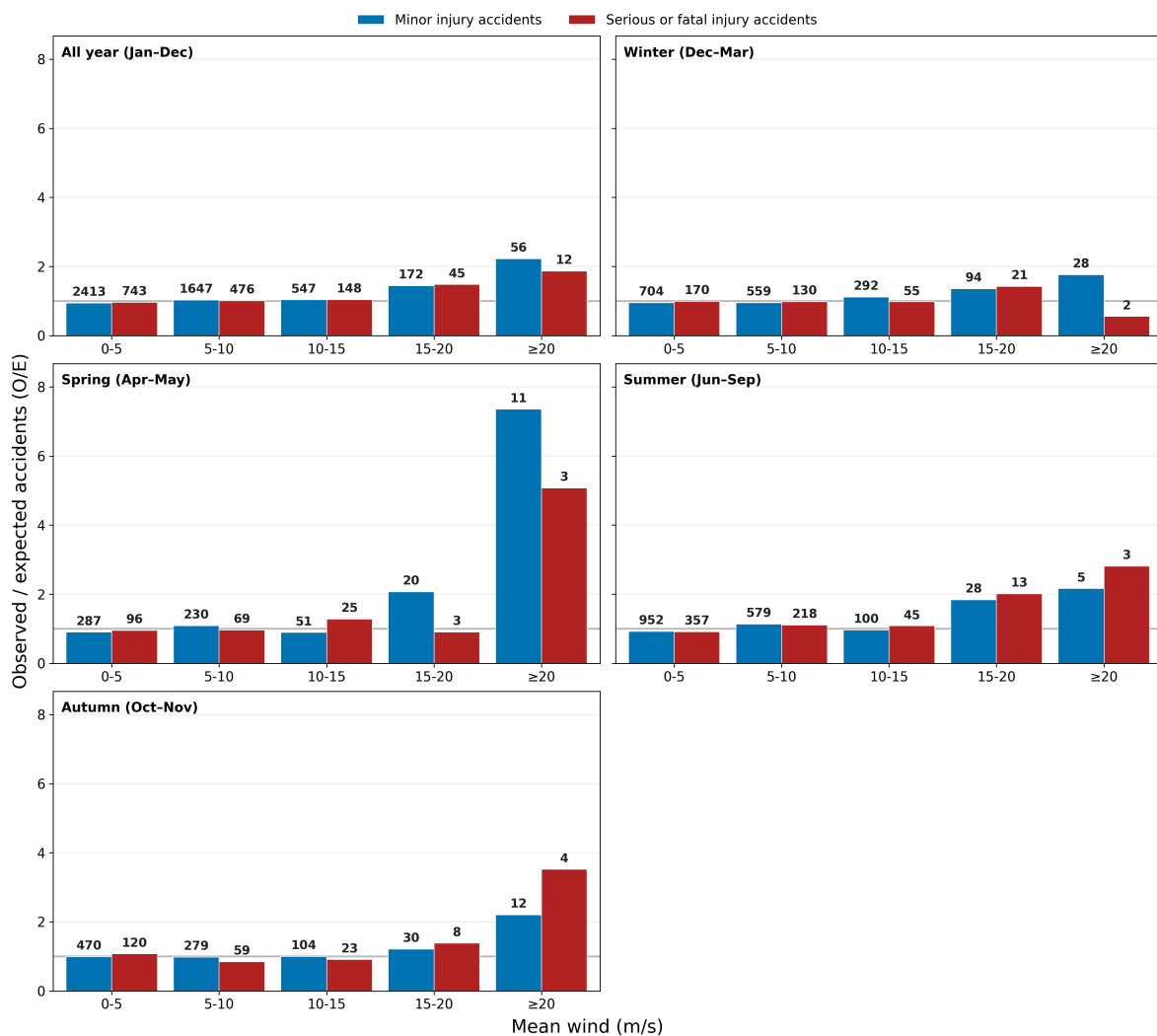


Figure 4.2: All-year and seasonal mean-wind O/E, 2007–2025. Blue bars show minor injury ( $meiðsli = 3$ ) and red bars show serious or fatal injury ( $meiðsli \leq 2$ ). Expected counts are calculated from local station–season weather frequency separately for each group; labels above bars are observed accident counts. The grey reference line is  $O/E = 1$ . Seasonal upper-wind bars should be interpreted with their small observed counts.

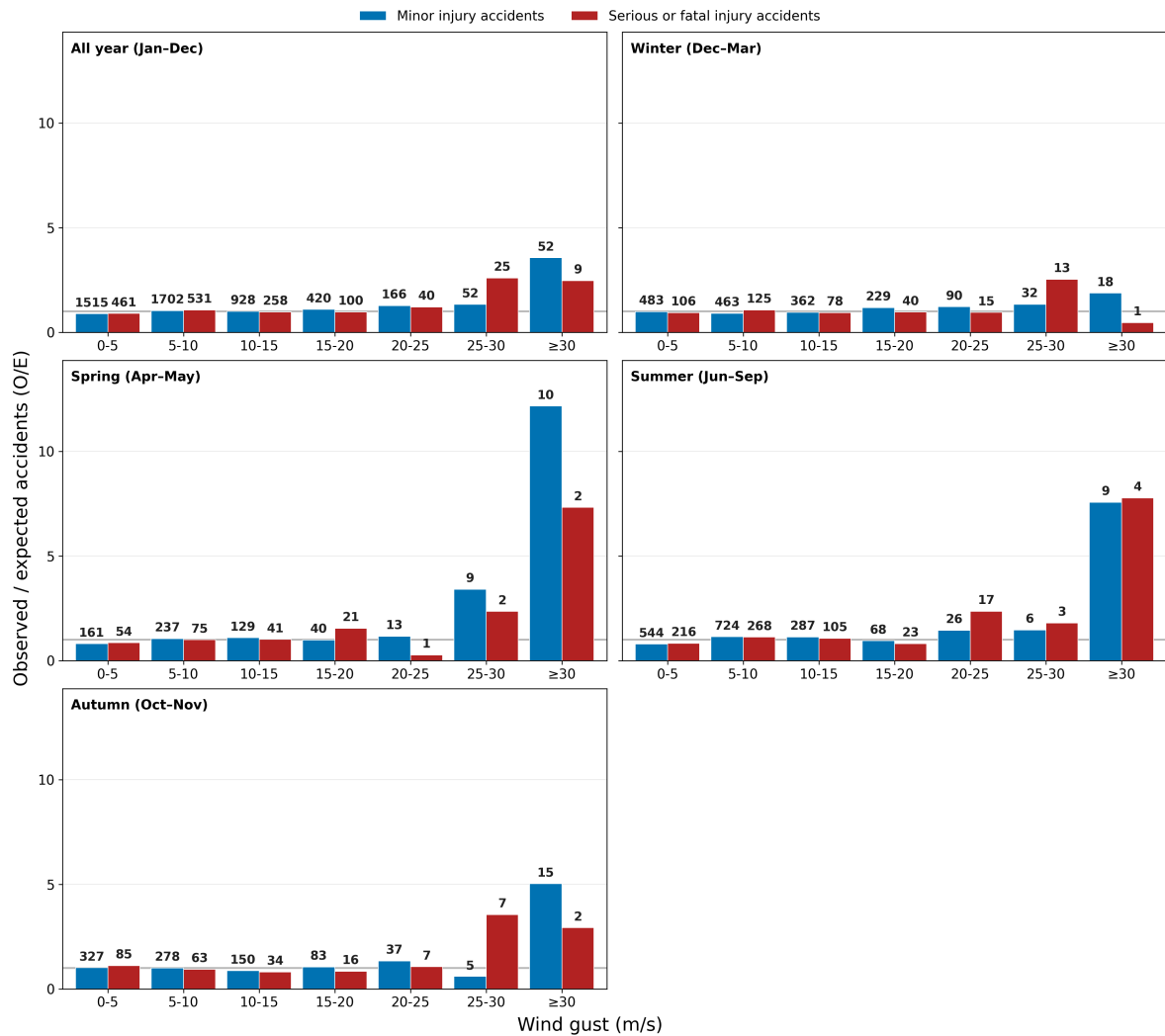


Figure 4.3: All-year and seasonal wind-gust O/E. Gust is taken from the same 10-minute observation as the accident-time mean wind and is not a daily maximum. Blue bars show minor injury and red bars show serious or fatal injury; labels above bars are observed accident counts. The grey reference line is  $O/E = 1$ . Several seasonal upper-gust cells contain few accidents and are therefore descriptive.

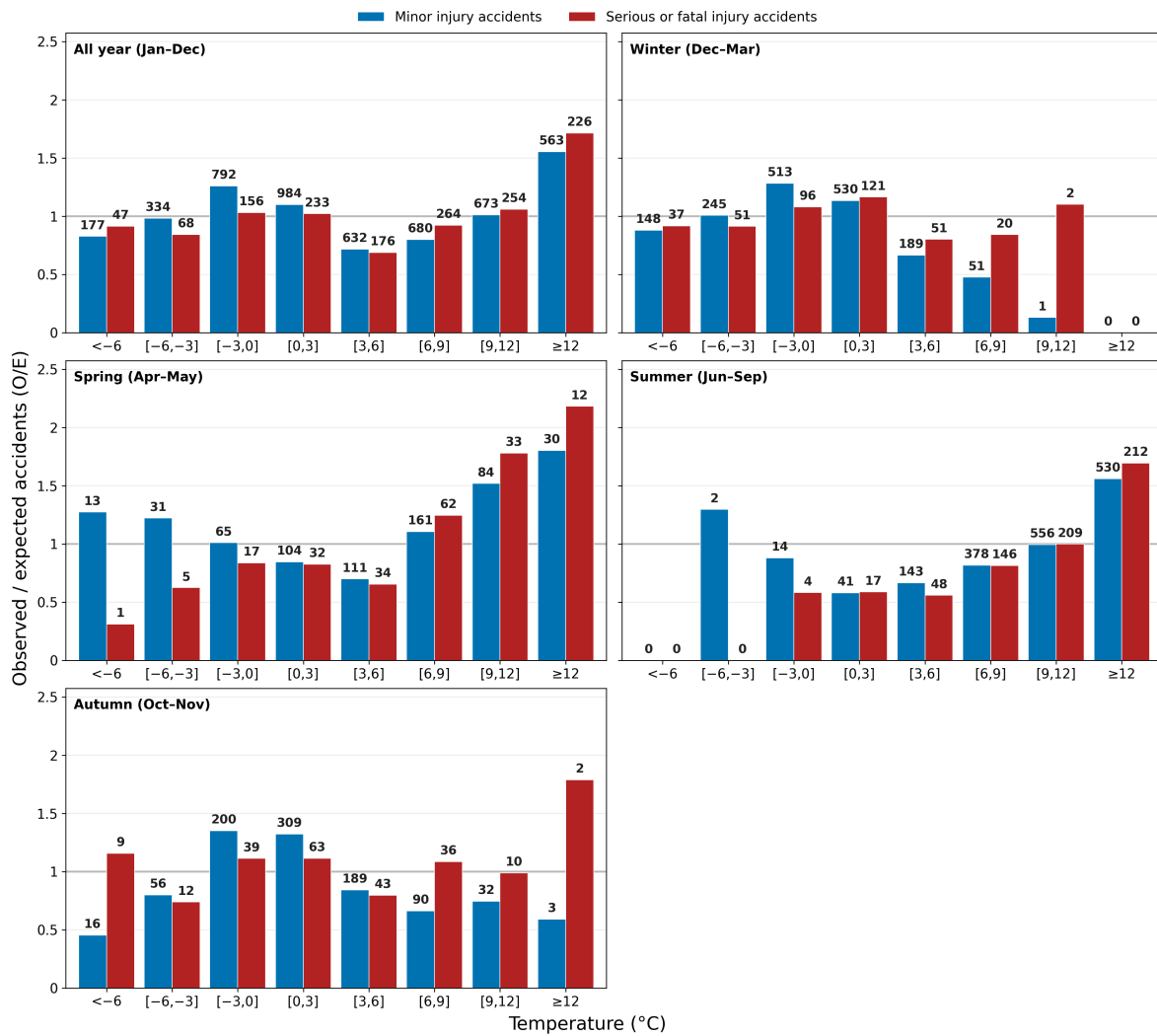


Figure 4.4: All-year and seasonal temperature O/E, standardised within matched temperature station and study season. Blue bars show minor injury and red bars show serious or fatal injury; labels above bars are observed accident counts. The grey reference line is O/E = 1. The non-monotonic pattern and sparse seasonal extremes are interpreted cautiously.

### 4.3 Q2: Approximate Traffic Correction to O/E

The primary O/E uses weather time rather than vehicle exposure. Daily-counter data from 2019–2024 show that traffic is lower in the strongest wind and gust intervals, so Q2 applies the estimated counter response to the primary station–season denominator as a sensitivity analysis. The observed accident counts remain unchanged; only the expected distribution is reweighted.

For mean wind of at least 20 m/s, the correction reduces the expected count from 31.6 to 25.1 and changes O/E from 2.15 to 2.71. For gusts of at least 30 m/s, expected accidents fall from 18.2 to 14.5 and O/E changes from 3.34 to 4.20. Figure 4.5 shows the corrected values for the same broad 2007–2025 accident population used in Q1.

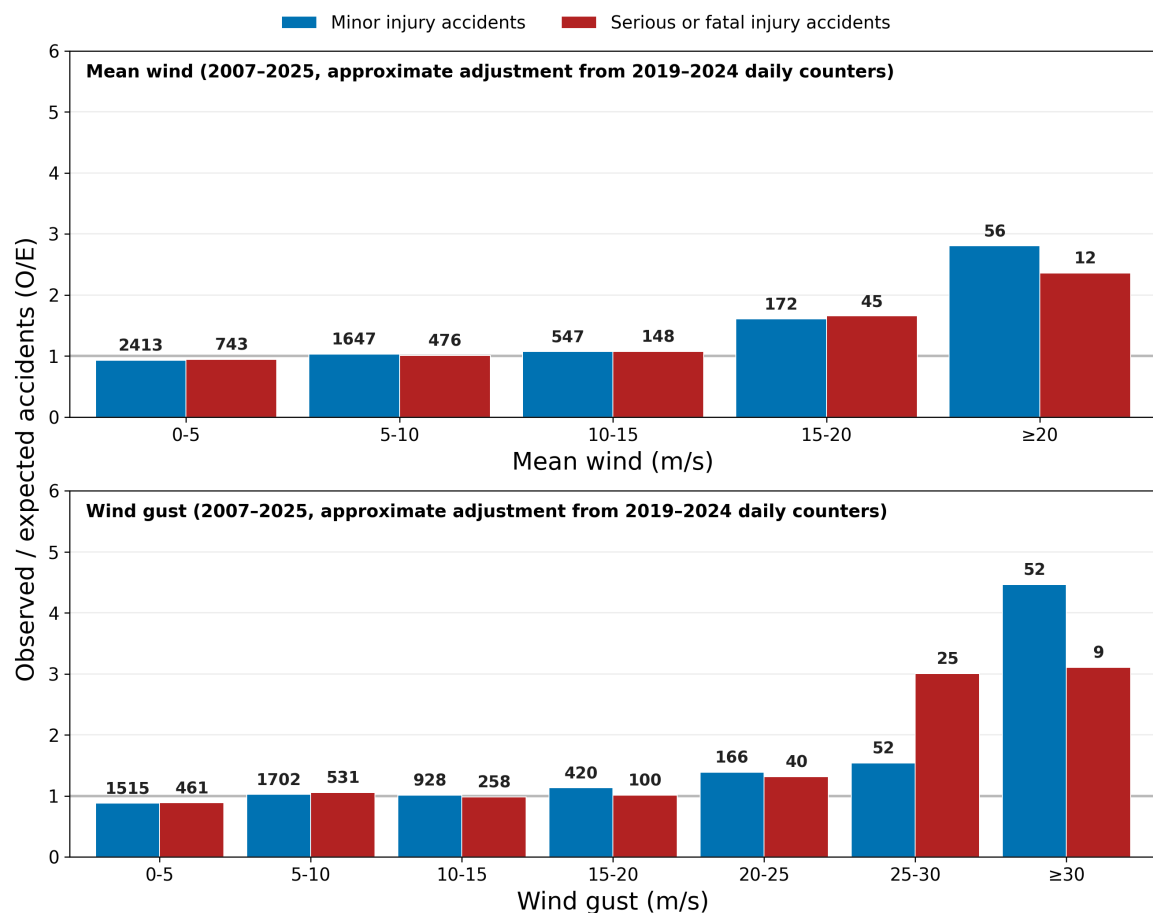


Figure 4.5: Approximate traffic-corrected O/E for the 2007–2025 primary sample. Observed counts use weather at each accident time. Expected counts start from the Q1 station–season weather frequencies and are reweighted using the daily traffic response estimated from counters in 2019–2024. The bars remain separated by injury group; the combined all-injury values are 2.71 for mean wind at  $\geq 20$  m/s and 4.20 for gust at  $\geq 30$  m/s. Because the counter response is extrapolated to other roads and years and daily totals do not provide hourly traffic, this is an approximate sensitivity analysis rather than a directly observed traffic-adjusted accident rate.

The correction therefore increases the upper-wind contrast under the assumption that the traffic response observed at the selected counters is informative for the wider road network.

It does not establish the magnitude of a national per-vehicle-kilometre contrast, because the traffic response is estimated over a shorter period and from a geographically selective set of counters.

## 4.4 Q3: Monthly-Frequency Daily-Counter Rate

Question 3 restricts the analysis to accidents and roads that can be linked to daily traffic counts. Its denominator contains 533,649 eligible rural counter-section days and 5.630 billion estimated vehicle-kilometres, including accident-free days. The 694 selected accidents are classified by weather at the recorded accident time, whereas each day's traffic exposure is distributed across weather intervals using the pooled calendar-month weather frequency at the station assigned to that counter section. Table 4.3 reports the resulting accident counts, estimated exposure and rates.

*Table 4.3: Monthly-frequency daily-counter rates, 2019–2024. Estimated exposure equals the observed daily traffic count times rural section length, allocated across weather intervals using the assigned station's pooled 2007–2025 calendar-month weather frequency for 07:00–24:00.*

| Measure   | Interval (m/s) | Accidents | Million VKT | Rate per million VKT |
|-----------|----------------|-----------|-------------|----------------------|
| Mean wind | 0–5            | 344       | 3011.5      | 0.114                |
| Mean wind | 5–10           | 210       | 1963.5      | 0.107                |
| Mean wind | 10–15          | 96        | 539.9       | 0.178                |
| Mean wind | 15–20          | 33        | 98.9        | 0.334                |
| Mean wind | ≥20            | 11        | 16.5        | 0.667                |
| Gust      | 0–5            | 205       | 1840.9      | 0.111                |
| Gust      | 5–10           | 238       | 2133.5      | 0.112                |
| Gust      | 10–15          | 129       | 1085.1      | 0.119                |
| Gust      | 15–20          | 70        | 400.3       | 0.175                |
| Gust      | 20–25          | 29        | 123.3       | 0.235                |
| Gust      | 25–30          | 12        | 34.5        | 0.348                |
| Gust      | ≥30            | 11        | 12.7        | 0.866                |

The resulting rates increase in the upper wind intervals. At mean wind of at least 20 m/s, 11 accidents occur in 16.5 million estimated VKT, corresponding to 0.667 accidents per million VKT. The 0–5 m/s rate is 0.114, so the descriptive rate ratio is 5.83. For gusts of at least 30 m/s, the corresponding rate is 0.866 per million VKT, 7.78 times the 0–5 m/s rate. Both upper intervals contain only 11 accidents, which limits the precision of the large ratios. The geographic selectivity of the counter-linked sample also limits their generalisability. Figure 4.6 shows the corresponding rate pattern across all wind and gust intervals.

A stratified diagnostic gives smaller upper-bin contrasts when the comparison is made within counter-section–year–season strata: O/E is 3.39 for mean wind and 4.84 for gust. This suggests that sample and road-exposure composition contribute to the larger pooled VKT rate ratios. The interpretation of this difference is considered in Chapter 5 rather than treated as a separate main result here.

Table 4.4 places the headline mean-wind results from Q1–Q3 beside the matched-time

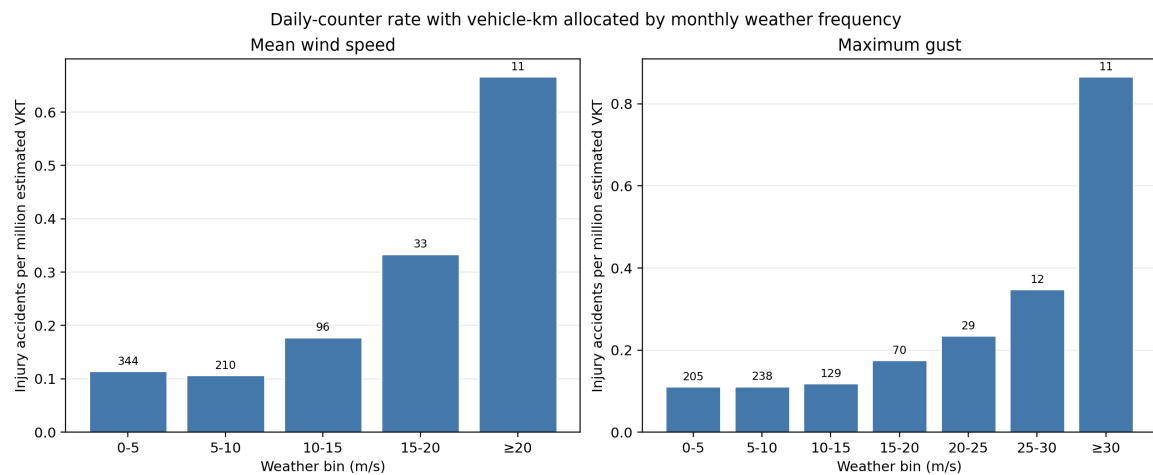


Figure 4.6: Rural injury accidents per million estimated vehicle-kilometres in the 694-accident counter-linked sample, 2019–2024. The denominator includes every eligible counter-section day, including days without accidents. The observed 24-hour traffic count is multiplied by rural road length and allocated across wind or gust intervals using the assigned station’s pooled 2007–2025 calendar-month weather frequency for 07:00–24:00. Accident labels above bars are counts classified by observed event-time weather. The monthly allocation and absence of hourly traffic are modelling assumptions.

robustness estimate. The entries are intended as an interpretation guide rather than a numerical ranking because the analyses use different samples, denominators, and estimands.

Table 4.4: Headline results from the three main wind analyses. These estimates have different denominators and should not be compared as repeated estimates of one parameter.

| Analysis                           | Headline estimate                                          | Interpretation                                                              |
|------------------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------|
| Q1: weather-frequency O/E          | ≥ 20 m/s: O/E 2.15                                         | Primary result; local weather-frequency denominator, no traffic adjustment. |
| Q2: approximate traffic correction | ≥ 20 m/s: O/E 2.71                                         | Q1 denominator reweighted by the 2019–2024 counter traffic response.        |
| Q3: monthly-frequency VKT          | ≥ 20 m/s: 0.667 vs 0.114 accidents per million VKT (5.83×) | Counter-linked 2019–2024 sample; 11 upper-bin accidents.                    |
| Supporting matched-time            | ≥ 15 vs 0–5 m/s: OR 1.61 (1.38–1.87)                       | Same station and recurring calendar time.                                   |

## 4.5 Supporting Analyses

The remaining analyses test specific aspects of the main Q1–Q3 pattern. They are reported as supporting checks because they either use different conditioning, smaller selected samples, or additional exposure assumptions.

### 4.5.1 Matched-Time and Secondary Weather Checks

The time-stratified case-crossover analysis contains 6,257 accident strata for each weather measure. Table 4.5 summarises the main categorical contrasts. Mean wind of at least 15 m/s has an odds ratio of 1.61 (95% CI 1.38–1.87) relative to 0–5 m/s, while lower and moderate wind contrasts are near one. This supports the upper-wind pattern after controlling for station, year, month, weekday and clock time, but it does not control unusual traffic or road conditions on a particular day.

*Table 4.5: Selected matched-time contrasts. Each model uses 6,257 accident strata; the joint wind contrast includes temperature. Intervals are 95% confidence intervals.*

| Comparison                                        | OR   | 95% CI    |
|---------------------------------------------------|------|-----------|
| Mean wind: $\geq 15$ vs 0–5 m/s                   | 1.61 | 1.38–1.87 |
| Mean wind, with temperature: $\geq 15$ vs 0–5 m/s | 1.68 | 1.44–1.96 |
| Gust: $\geq 30$ vs 0–10 m/s                       | 2.74 | 1.96–3.85 |
| Temperature: $-3-0$ vs $0-3$ °C                   | 1.13 | 1.02–1.25 |
| Temperature: $3-6$ vs $0-3$ °C                    | 0.59 | 0.53–0.65 |
| Temperature: $\geq 15$ vs $0-3$ °C                | 0.99 | 0.80–1.23 |

The gust model also rises in the upper intervals, whereas the temperature results remain non-monotonic. Including temperature in the mean-wind model changes the upper-wind estimate only slightly, from OR 1.61 to 1.68. The matched-time temperature result reproduces the dip at moderate temperatures but not the warm-end increase seen in Q1, reinforcing the cautious interpretation of temperature.

Daylight and severity provide additional context rather than separate main findings. The matched daylight estimates are imprecise because only 985 matched sets change daylight class. In the adjusted severity model, mean wind of at least 15 m/s has OR 0.93 (95% CI 0.69–1.26) relative to 0–10 m/s, providing no clear evidence that strong wind is associated with greater severity once an injury accident has occurred. The full severity model is reported in Table A.1 in the appendix. A matched-time wind-by-season interaction test gives  $\chi^2(6) = 14.30$ ,  $p = 0.026$ , but sparse upper-wind cells do not support ranking individual seasons.

### 4.5.2 Supporting Traffic Models

The additional traffic models provide checks using different exposure constructions. The broad annual model retains 5,125 accidents and compares rates within road section, year and traffic period. Its rate ratio is 2.34 (95% CI 1.72–3.17) for 20–25 versus 0–5 m/s and rises further in the sparse  $\geq 25$  m/s interval. These estimates are consistent in direction with Q1–Q3, but their magnitude is not directly comparable because seasonal traffic is allocated by weather frequency and the linked road sample differs.

A separate allocated daily-counter model retains 762 accidents and gives RR 3.62 (95% CI 2.65–4.94) for  $\geq 15$  versus 0–10 m/s. It uses observed daily totals but an estimated within-day allocation and is therefore retained as a robustness check rather than a main result. Appendix Table A.5 reports the principal traffic sensitivity checks, including the official VDU/SDU restriction and the effect of excluding recorded zero counter-days.

Across the primary and supporting analyses, the common feature is an elevation in the upper wind intervals. The analyses differ in population, denominator and estimand, so agreement in direction is more informative than comparison of their numerical magnitudes.



# 5 Discussion

## 5.1 Interpretation

The three main analyses address the wind result using different levels of traffic information and different exposure assumptions. Q1 shows that rural injury accidents are over-represented during strong mean wind relative to how often those conditions occur locally: at  $\geq 20$  m/s, the all-injury O/E is 2.15. Q2 applies an approximate traffic correction derived from daily counters and increases the same upper-bin O/E to 2.71. Q3 uses observed daily traffic and rural road length in a smaller counter-linked sample and gives a rate at  $\geq 20$  m/s that is 5.83 times the 0–5 m/s rate. The three values are not estimates of the same quantity, but each places the strongest contrast in the upper wind intervals.

Q2 addresses an important limitation of weather-frequency O/E: traffic volume is not constant across weather conditions. At the selected counters, daily traffic is lower during stronger wind. If that response also applies to the broader rural road network, weighting expected accidents by the observed traffic response makes the upper-wind contrast larger rather than explaining it away. The correction remains approximate, however. It extrapolates a 2019–2024 counter relationship to the 2007–2025 primary population, and it uses daily traffic totals rather than traffic observed in the same 10-minute intervals used to classify accidents. Q2 is therefore best interpreted as a sensitivity analysis of the primary O/E rather than as a measured traffic-adjusted accident rate.

Q3 incorporates day-to-day traffic variation directly in the denominator, but its larger rate ratio requires separate interpretation. The 694-accident sample covers only roads and years with usable counter information, and only 11 accidents occur in the  $\geq 20$  m/s mean-wind interval. In addition, pooled vehicle-kilometre rates can be influenced by which road sections contribute exposure to each wind interval. A diagnostic comparison within counter-section-year-season strata gives upper-bin O/E 3.39 for mean wind and 4.84 for gust, smaller than the pooled Q3 rate ratios. This suggests that differences in road and exposure composition account for part of the larger pooled contrasts. It does not establish that particular roads are intrinsically more dangerous, and the sparse upper bins leave substantial uncertainty about the exact magnitude of the contrast.

The supporting analyses are consistent with this interpretation but answer different questions. The matched-time comparison retains an upper-wind association after controlling for station and recurring calendar time. This makes an explanation based solely on regular differences between months, weekdays or times of day less plausible. Gust shows a similar upper-tail pattern, although mean wind and gust are related measurements from the same weather observations. Temperature is less consistent: its O/E pattern is non-monotonic and the matched-time analysis does not reproduce the warm-end increase. The severity model likewise gives no clear evidence that strong wind is associated with a greater probability of serious or fatal injury once an injury accident has occurred. These results support a distinction

between the occurrence of accidents and their severity.

The main wind result is consistent with earlier work comparing accident shares with the time spent in adverse weather (Edwards, 1994) and with engineering research showing mechanisms by which crosswind can impair vehicle stability (Baker, 1986, 1994). The contribution here is not a new physical wind threshold, but a long Icelandic accident series linked to local weather frequency and then examined under several complementary exposure constructions. The non-monotonic temperature findings are also consistent with previous evidence that temperature associations can vary by range and season (Lee et al., 2014; Zhan et al., 2020), while the observed reduction in traffic during strong wind agrees with recorder-based evidence that adverse weather changes travel demand (Sathiaraj et al., 2018).

## 5.2 Strengths

A principal strength is the broad primary sample: 6,259 rural injury accidents are linked to quality-controlled 10-minute weather observations, and expected counts are standardised within weather station and study season rather than against a single national weather distribution. Match coverage, observed counts and weather-quality exclusions are reported explicitly, which makes the primary O/E calculation transparent and auditable.

The exposure analyses complement rather than replace this broad comparison. Q2 tests how the primary result changes under an empirically estimated traffic response, while Q3 uses observed daily traffic, road length and all eligible counter-days, including days without an accident. The matched-time and additional traffic models provide supporting checks under different conditioning and exposure assumptions. Keeping these analyses separate makes their assumptions visible instead of combining incompatible data sources into one apparently precise measure. Documented transformations, automated tests and stage-specific audits further support reproducibility.

## 5.3 Limitations

The main limitation is incomplete traffic exposure. Q1 uses weather time rather than vehicle-kilometres. Q2 introduces traffic only through an approximate response estimated from selected counters in 2019–2024 and extrapolated to the broader 2007–2025 population. The counter data are daily totals, so neither Q2 nor Q3 observes how many vehicles were travelling during the specific 10-minute weather interval in which an accident occurred.

Q3 is more exposure-based but also more selective. It is limited to rural counter sections with usable traffic, road geometry and weather in 2019–2024. The full observed daily count is allocated to 07:00–24:00 using pooled station-month weather frequencies, an assumption that preserves daily exposure but does not reconstruct hourly traffic. The monthly weather frequencies use the longer 2007–2025 weather record for stability, while the traffic and accidents themselves are restricted to 2019–2024. Only 11 Q3 accidents occur in each of the highest mean-wind and gust intervals, so the corresponding large rate ratios should be interpreted as directional evidence rather than precise estimates. An assigned counter also represents exposure on a road section; it does not establish that the vehicle involved in an

accident passed that counter.

Weather measurement introduces a separate source of uncertainty. A station up to 20 km from an accident or counter location is a proxy for conditions on the road, and Icelandic terrain can produce substantial variation over shorter distances. The five-minute matching rule can also select an observation on the adjacent calendar day for accidents close to midnight; this preserves the time tolerance but should be understood as part of the matching definition. The fixed 2020–2024 urban-boundary dataset is applied to the full 2007–2025 accident period, so historical changes in settlement boundaries are not represented.

The supporting models have their own restrictions. The case-crossover design controls recurring calendar time but not unusual traffic, road-surface conditions, visibility or closures on a particular day. The annual traffic model distributes seasonal traffic across wind intervals using weather frequency rather than observed wind-specific traffic. More generally, all analyses are observational. Unmeasured road characteristics, vehicle type, speed, visibility, driver behaviour and weather-related road management may contribute to the observed patterns. The study therefore supports associations and exposure contrasts, not causal effects of wind.

## **5.4 Future Work**

The most valuable extension would be verified hourly traffic linked to stable counter coordinates, road-section identifiers, direction information and counter downtime. Hourly exposure would remove the need to distribute daily traffic totals across weather intervals and would allow accident-time weather to be paired with traffic measured at a compatible temporal resolution. A second useful extension would examine baseline accident rates and exposure composition across road sections to determine how much of the difference between pooled VKT rates and locally standardised O/E is attributable to road mix. With sufficiently detailed data, vehicle class, road geometry and wind direction relative to the road could also be incorporated.



## 6 Conclusion

The primary weather-frequency analysis identifies a clear upper-wind pattern in rural injury accidents in Iceland. Of 6,414 rural injury accidents in 2007–2025, 6,259 have qualifying weather matches within 20 km and five minutes. At mean wind speeds of at least 20 m/s, 68 accidents were observed compared with 31.6 expected from local weather frequency, giving O/E 2.15. This is the main result of the thesis: strong wind is over-represented at accident times relative to how often it occurs locally.

The traffic analyses refine the interpretation while also defining its limits. Reweighting the primary expected counts with the traffic response observed at counters increases the upper-wind O/E from 2.15 to 2.71, but this is an approximate extrapolation rather than a measured traffic-adjusted rate. In the separate 694-accident monthly-frequency counter analysis, the  $\geq 20$  m/s rate is 0.667 accidents per million estimated VKT, 5.83 times the 0–5 m/s rate. Only 11 accidents occur in this upper interval, and part of the larger pooled contrast is related to road and exposure composition.

Supporting matched-time and traffic models place the strongest contrasts in the same upper-wind range, while the temperature results are less consistent and no clear increase in injury severity is evident in strong wind. Taken together, the analyses show that rural injury accidents are relatively more frequent during strong mean wind, but they do not establish a causal effect or a complete national accident risk per vehicle-kilometre. Estimating such a quantity directly would require wider geographic coverage and traffic measured at an hourly or finer temporal resolution.



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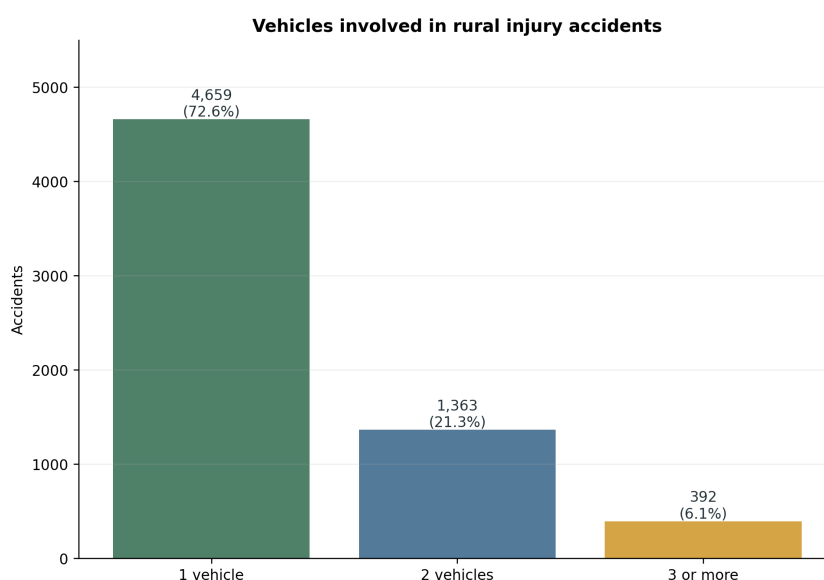
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# A Supporting Checks

This appendix contains the checks most relevant to interpreting the main results. Additional checks are documented separately.

## A.1 Accident Characteristics

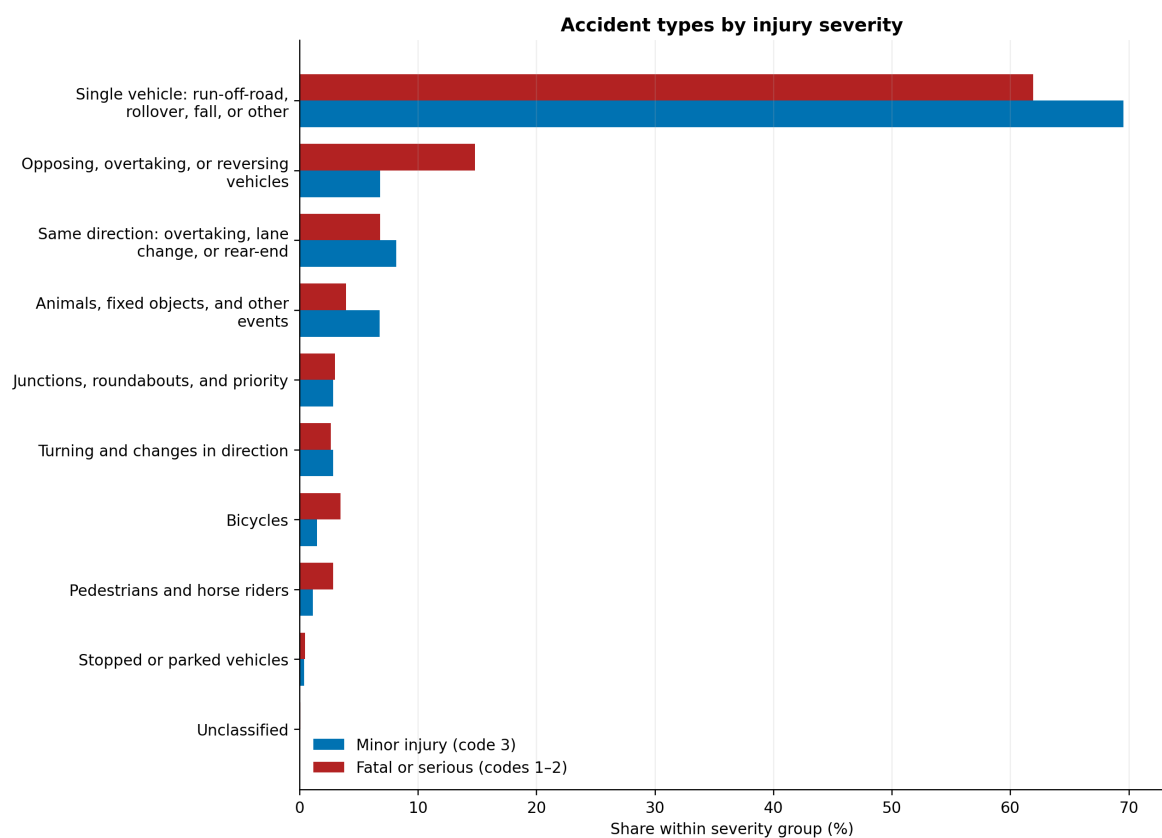
Figures A.1 and A.2 complement the hour, season and daylight counts in the data chapter. They describe the selected injury accidents without a traffic-exposure denominator.



*Figure A.1: Number of registered vehicles involved in the 6,414 rural injury accidents. Counts describe the selected sample, not risk by vehicle count.*

## A.2 Adjusted Severity Model

Table A.1 shows the adjusted comparison of serious or fatal and minor-injury accidents. It describes severity among recorded injury accidents; it does not estimate accident occurrence or risk per vehicle-kilometre.



*Figure A.2: Accident-type composition for minor injury (blue, code 3) and serious or fatal injury (red, codes 1–2). Percentages sum to 100 within each separate severity group.*

*Table A.1: Adjusted odds ratios for a serious or fatal outcome among recorded injury accidents. All listed variables are included in the same model.*

| Variable    | Comparison                 | Adjusted OR (95% CI) |
|-------------|----------------------------|----------------------|
| Mean wind   | 10–15 m/s vs 0–10 m/s      | 0.99 (0.82–1.21)     |
| Mean wind   | $\geq 15$ m/s vs 0–10 m/s  | 0.93 (0.69–1.26)     |
| Temperature | $< -6$ °C vs 0–3 °C        | 1.13 (0.79–1.62)     |
| Temperature | $-6$ – $-3$ °C vs 0–3 °C   | 0.85 (0.63–1.15)     |
| Temperature | $-3$ –0 °C vs 0–3 °C       | 0.83 (0.66–1.04)     |
| Temperature | 3–6 °C vs 0–3 °C           | 1.11 (0.88–1.39)     |
| Temperature | 6–9 °C vs 0–3 °C           | 1.41 (1.10–1.79)     |
| Temperature | 9–12 °C vs 0–3 °C          | 1.28 (0.98–1.68)     |
| Temperature | 12–15 °C vs 0–3 °C         | 1.33 (0.98–1.81)     |
| Temperature | $\geq 15$ °C vs 0–3 °C     | 1.21 (0.83–1.76)     |
| Daylight    | Darkness vs Daylight       | 0.89 (0.71–1.12)     |
| Daylight    | Civil twilight vs Daylight | 1.09 (0.87–1.36)     |
| Time of day | 00–06 vs 10–16             | 0.86 (0.64–1.16)     |
| Time of day | 06–10 vs 10–16             | 0.65 (0.52–0.82)     |
| Time of day | 16–20 vs 10–16             | 0.98 (0.84–1.14)     |
| Time of day | 20–24 vs 10–16             | 0.77 (0.61–0.98)     |
| Season      | Winter vs Summer           | 0.82 (0.64–1.05)     |
| Season      | Spring vs Summer           | 0.96 (0.78–1.18)     |
| Season      | Autumn vs Summer           | 0.83 (0.65–1.05)     |

## A.3 Supporting Allocated Daily-Counter Sample

This section documents the supporting allocated daily-counter model, not the 694-accident Q3 monthly-frequency VKT analysis. The tables show how the 2019–2024 rural injury sample narrows under the linkage used by that supporting model. The 767 records at the last broad selection step include five records that do not enter its final allocated-rate model, which retains 762 accidents.

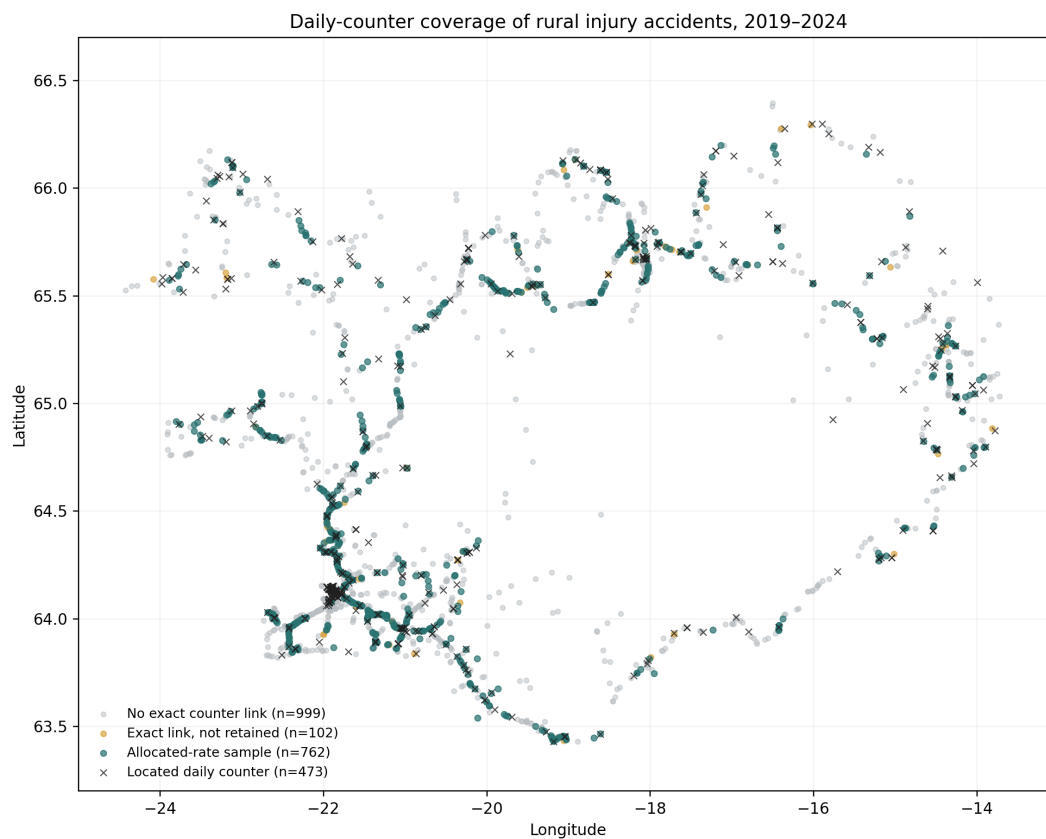
*Table A.2: Selection of accidents for the supporting allocated daily-counter model.*

| Selection step                             | Accidents | Share of 2019–2024 sample |
|--------------------------------------------|-----------|---------------------------|
| Rural injury accidents in 2019–2024        | 1,863     | 100.0%                    |
| Exact road-section/year counter candi-date | 864       | 46.4%                     |
| Assigned counter within 20 km              | 863       | 46.3%                     |
| Positive count and valid full-day wind     | 767       | 41.2%                     |

*Table A.3: Characteristics of accidents retained and excluded by the linkage used for the supporting allocated daily-counter model.*

| Characteristic         | All    | No link | Linked, excluded | Retained |
|------------------------|--------|---------|------------------|----------|
| Accidents              | 1,863  | 999     | 102              | 762      |
| Share of all accidents | 100.0% | 53.6%   | 5.5%             | 40.9%    |
| Distinct road sections | 563    | 374     | 56               | 189      |
| Serious or fatal       | 25.6%  | 29.4%   | 18.6%            | 21.5%    |
| One vehicle            | 67.7%  | 75.2%   | 54.9%            | 59.7%    |
| Winter                 | 32.6%  | 28.0%   | 40.2%            | 37.5%    |
| Spring                 | 13.0%  | 11.7%   | 8.8%             | 15.4%    |
| Summer                 | 36.0%  | 41.8%   | 33.3%            | 28.6%    |
| Autumn                 | 18.4%  | 18.4%   | 17.6%            | 18.5%    |

The geographic selectivity of this traffic-linked sample is documented in Figure A.3.



*Figure A.3: Geographic coverage of located daily counters and rural injury accidents retained or excluded by the daily-counter linkage, 2019–2024.*

## A.4 Traffic Sensitivity Checks

The assignment-distance check shows how the supporting allocated daily-counter result changes under 5, 10, and 20 km limits. It is retained to make the spatial selection rule explicit.

*Table A.4: Upper-bin estimate from the supporting allocated daily-counter model under three assignment distances.*

| Maximum distance | Included accidents | Upper-bin accidents | Rate ratio | 95% CI    |
|------------------|--------------------|---------------------|------------|-----------|
| 5 km             | 585                | 12                  | 2.91       | 1.58–5.37 |
| 10 km            | 750                | 14                  | 2.78       | 1.58–4.88 |
| 20 km            | 767                | 14                  | 2.73       | 1.55–4.78 |

Table A.5 combines the remaining traffic checks. It compares the annual model using all traffic periods with the version using only official VDU and SDU, reports the effect of excluding zero counter-days, and records exclusions caused by nonpositive VDU or derived spring/autumn daily traffic values.

*Table A.5: Sensitivity and quality checks for the supporting traffic analyses.*

| Check                     | Data included                           | Estimate (95% interval) | Records             |
|---------------------------|-----------------------------------------|-------------------------|---------------------|
| Rate model, 20–25 m/s     | All VDU, SDU, and derived VH DU periods | RR 2.34 (1.72–3.17)     | 44 accidents        |
| Rate model, 20–25 m/s     | Official VDU and SDU periods only       | RR 2.03 (1.38–2.96)     | 28 accidents        |
| Rate model, $\geq 25$ m/s | All VDU, SDU, and derived VH DU periods | RR 5.16 (3.15–8.44)     | 17 accidents        |
| Rate model, $\geq 25$ m/s | Official VDU and SDU periods only       | RR 2.84 (1.34–6.04)     | 7 accidents         |
| Daily traffic, 20–25 m/s  | Retain zero counter-days                | 68.94% (62.33–74.09)    | 1,175 days          |
| Daily traffic, 20–25 m/s  | Exclude zero counter-days               | 69.29% (62.70–74.52)    | 1,090 days          |
| Nonpositive published VDU | Excluded, not imputed                   | 6.57% excluded          | 1,509 section-years |
| Nonpositive derived VH DU | Excluded, not imputed                   | 2.40% excluded          | 552 section-years   |